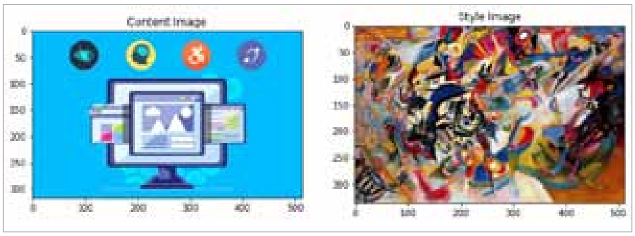

Fig. 1: A source image and a style image

Fig. 2: Fast style transfer

- Extracting the style and content
- Running the gradient descent
- Total variation loss
- Re-running the optimisation

The final stylised output with the source image rendered using the reference style image is shown in Fig. 4.

The complete code demo is available at *https://colab.research.google.com/github/tensorflow/docs/blob/master/site/en/tutorials/generative/style_transfer.ipynb*

MAGENTA

Magenta is an open source project aimed at exploring the role of machine learning in the creative process *(https://magenta.tensorflow.org/)*. There are two different distributions of Magenta, as listed below.

Magenta-Python. This is a Python library that is built on TensorFlow. It has powerful methods to manipulate music and image data. Generating music by using ML models is the primary objective of Magenta-Python. It can be installed with pip.

Magenta-JavaScript. This distribution is an open source API. It enables the use of pre-trained Magenta models inside the Web browser. TensorFlow.js is at the core of this API.

Magenta-Python

The first step is to install Magenta along with the dependencies, as follows:

```
print('Installing dependencies...')
!apt-get update -qq && apt-get install -qq libfluidsynth1
fluid-soundfont-gm build-essential libasound2-dev libjack-dev
!pip install -qU pyfluidsynth pretty_midi
!pip install -qU magenta
```

Magenta fundamentally works on NoteSequences, which are abstract representations of a sequence of notes. For example, the following code snippet represents 'Twinkle Twinkle Little Star':

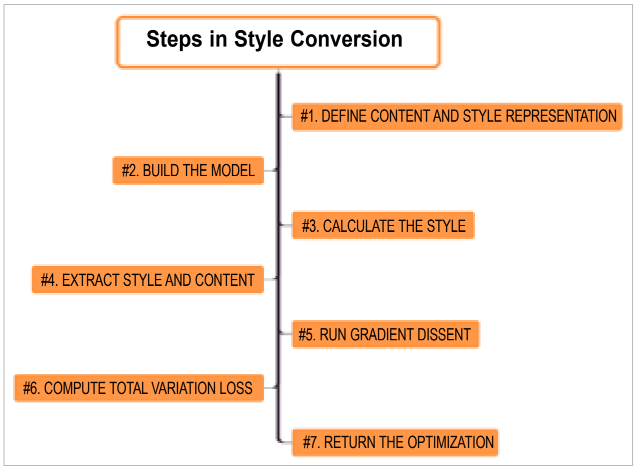

Fig. 3: Steps involved in the styling process

Fig. 4: Stylised output

```
from magenta.music.protobuf import music_pb2
twinkle_twinkle = music_pb2.NoteSequence()
# Add the notes to the sequence.
twinkle_twinkle.notes.add(pitch=60, start_time=0.0, end_time=0.5,
velocity=80)
twinkle_twinkle.notes.add(pitch=60, start_time=0.5, end_time=1.0,
velocity=80)
twinkle_twinkle.notes.add(pitch=67, start_time=1.0, end_time=1.5,
velocity=80)
twinkle_twinkle.notes.add(pitch=67, start_time=1.5, end_time=2.0,
velocity=80)
twinkle_twinkle.notes.add(pitch=69, start_time=2.0, end_time=2.5,
velocity=80)
twinkle_twinkle.notes.add(pitch=69, start_time=2.5, end_time=3.0,
velocity=80)
twinkle_twinkle.notes.add(pitch=67, start_time=3.0, end_time=4.0,
velocity=80)
twinkle_twinkle.notes.add(pitch=65, start_time=4.0, end_time=4.5,
velocity=80)
twinkle_twinkle.notes.add(pitch=65, start_time=4.5, end_time=5.0,
velocity=80)
twinkle_twinkle.notes.add(pitch=64, start_time=5.0, end_time=5.5,
velocity=80)
twinkle_twinkle.notes.add(pitch=64, start_time=5.5, end_time=6.0,
velocity=80)
twinkle_twinkle.notes.add(pitch=62, start_time=6.0, end_time=6.5,
velocity=80)
twinkle_twinkle.notes.add(pitch=62, start_time=6.5, end_time=7.0,
```